

DSHW4 Shortest Path

- ▶ Prim Algorithm
- ▶ Kruskal Algorithm
- ▶ Dijkstra Algorithm

Prim's Algorithm

- ▶ 實作 Prim's Algorithm
- ▶ 輸入為共有幾組Edge(第一行) , 多組 Edge 資訊 (vertex 1, vertex 2, weight) , 與起始點(最後一行)
- ▶ 輸出依 weight 升冪排列 , 兩個vertex按照字母順序

Sample Input 1 

```
11
a b 2
a d 9
b c 3
b d 8
b e 7
c e 4
d e 5
d f 10
e f 12
e g 11
f g 13
a
```

Sample Output 1

```
a b 2
b c 3
c e 4
d e 5
d f 10
e g 11
```

Kruskal's Algorithm

- ▶ 實作 Kruskal's Algorithm
- ▶ 輸入為共有幾組Edge(第一行) , 多組 Edge 資訊 (vertex 1, vertex 2, weight)
- ▶ 輸出依 weight 升冪排列 , 兩個vertex按照字母順序

Sample Input 1

```
9
a b 7
a c 6
a d 1
b c 8
c d 5
c e 3
d e 4
d f 5
e f 2
```

Sample Output 1

```
a d 1
e f 2
c e 3
d e 4
a b 7
```

Dijkstra's Algorithm

- ▶ 實作 Dijkstra's Algorithm
- ▶ 輸入為共有幾組Edge(第一行)，多組 Edge 資訊 (vertex 1, vertex 2, weight) ，與起始點(最後一行)
- ▶ Graph為有向圖，vertex 1為 Edge 起始點
- ▶ 輸出為起始點可到達的點與其所需的 cost，依 cost 升冪排列

Sample Input 1

```
9
b a 4
a c 2
f c 9
d b 15
d f 5
d g 23
e b 17
e g 11
f g 13
d
```

Sample Output 1

```
d 0
f 5
c 14
b 15
g 18
a 19
```

- ▶ 1.作業要上傳OJ並 submit 通過
- ▶ 2.繳交source code 與心得到moodle
- ▶ OJ預計在今晚會將題目開啟
- ▶ OJ網址 <http://nlpoj.ddns.net/>